

IDS Hardware Node: Wiring & Assembly Guide

This document details the physical I/O pin mappings required to build the ESP32 Security Operations Center (SOC) node.

ESP32 Pin	Component	Component Pin
3V3	OLED Display	VCC
GND	OLED Display	GND
D21	OLED Display	SDA
D22	OLED Display	SCL
D18	Green LED	Anode (+)
D5	Orange/Red LED	Anode (+)
GND	Both LEDs	Cathode (-)

Step-by-Step Assembly Instructions :

1. Wiring the OLED Screen (I2C)

The OLED screen uses the I2C protocol, which only requires 4 wires. The ESP32 has default hardware pins for this (21 and 22), so no extra code definition is needed for the data lines.

- Connect the **VCC** pin on the OLED to the **3V3** pin on the ESP32.
- Connect the **GND** pin on the OLED to a **GND** pin on the ESP32.
- Connect the **SDA** pin on the OLED to pin **D21** on the ESP32.
- Connect the **SCL** pin on the OLED to pin **D22** on the ESP32.

2. Wiring the Status LEDs

- **Green LED (SAFE):** Connect the long leg (Anode) to pin **D18**. Connect the short leg (Cathode) to a 330Ω resistor, and connect the other end of the resistor to **GND**.
- **Orange/Red LED (THREAT):** Connect the long leg (Anode) to pin **D5**. Connect the short leg (Cathode) to a 1000Ω resistor, and connect the other end of the resistor to **GND**.